

1.2 For each of these equalities, you must show containment in both directions.

- $x \in A \setminus B \Leftrightarrow x \in A$ and $x \notin B \Leftrightarrow x \in A$ and $x \notin A \cap B \Leftrightarrow x \in A \setminus (A \cap B)$. Also, $x \in A$ and $x \notin B \Leftrightarrow x \in A$ and $x \in B^c \Leftrightarrow x \in A \cap B^c$.
- Suppose $x \in B$. Then either $x \in A$ or $x \in A^c$. If $x \in A$, then $x \in B \cap A$, and, hence $x \in (B \cap A) \cup (B \cap A^c)$. Thus $B \subset (B \cap A) \cup (B \cap A^c)$. Now suppose $x \in (B \cap A) \cup (B \cap A^c)$. Then either $x \in (B \cap A)$ or $x \in (B \cap A^c)$. If $x \in (B \cap A)$, then $x \in B$. If $x \in (B \cap A^c)$, then $x \in B$. Thus $(B \cap A) \cup (B \cap A^c) \subset B$. Since the containment goes both ways, we have $B = (B \cap A) \cup (B \cap A^c)$. (Note, a more straightforward argument for this part simply uses the Distributive Law to state that $(B \cap A) \cup (B \cap A^c) = B \cap (A \cup A^c) = B \cap S = B$.)
- Similar to part a).
- From part b).

$$A \cup B = A \cup [(B \cap A) \cup (B \cap A^c)] = A \cup (B \cap A) \cup A \cup (B \cap A^c) = A \cup [A \cup (B \cap A^c)] = A \cup (B \cap A^c).$$

- 1.3 a. $x \in A \cup B \Leftrightarrow x \in A$ or $x \in B \Leftrightarrow x \in B \cup A$
 $x \in A \cap B \Leftrightarrow x \in A$ and $x \in B \Leftrightarrow x \in B \cap A$.
- $x \in A \cup (B \cup C) \Leftrightarrow x \in A$ or $x \in B \cup C \Leftrightarrow x \in A \cup B$ or $x \in C \Leftrightarrow x \in (A \cup B) \cup C$.
 (It can similarly be shown that $A \cup (B \cup C) = (A \cup C) \cup B$.)
 $x \in A \cap (B \cap C) \Leftrightarrow x \in A$ and $x \in B$ and $x \in C \Leftrightarrow x \in (A \cap B) \cap C$.
 - $x \in (A \cup B)^c \Leftrightarrow x \notin A$ or $x \notin B \Leftrightarrow x \in A^c$ and $x \in B^c \Leftrightarrow x \in A^c \cap B^c$
 $x \in (A \cap B)^c \Leftrightarrow x \notin A \cap B \Leftrightarrow x \notin A$ and $x \notin B \Leftrightarrow x \in A^c$ or $x \in B^c \Leftrightarrow x \in A^c \cup B^c$.

Note: $A_1 = (\frac{1}{2}, 1)$

$$A_2 = (\frac{1}{4}, \frac{1}{2})$$

$$A_3 = (\frac{1}{6}, \frac{1}{3})$$

$$A_4 = (\frac{1}{8}, \frac{1}{4})$$

...

$$\bigcap_{n=1}^{\infty} A_n = \emptyset$$

Since there isn't even any overlap in A_1 and A_2 , it is clear that there is no number or set of numbers in all of the A_n .

$$\bigcup_{n=1}^{\infty} A_n = (0, \frac{1}{2}) \cup (\frac{1}{2}, 1)$$

Since neither A_1 nor A_2 contain the number $\frac{1}{2}$ and neither do any other A_n , so $\frac{1}{2}$ isn't included in any of the A_n .

1.11 We must verify each of the three properties in Definition 1.2.1.

- (1) The empty set $\emptyset \in \{\emptyset, S\}$. Thus $\emptyset \in \mathcal{B}$. (2) $\emptyset^c = S \in \mathcal{B}$ and $S^c = \emptyset \in \mathcal{B}$. (3) $\emptyset \cup S = S \in \mathcal{B}$.
- (1) The empty set $\emptyset$ is a subset of any set, in particular, $\emptyset \subset S$. Thus $\emptyset \in \mathcal{B}$. (2) If $A \in \mathcal{B}$, then $A \subset S$. By the definition of complementation, A^c is also a subset of S , and, hence, $A^c \in \mathcal{B}$. (3) If $A_1, A_2, \dots \in \mathcal{B}$, then, for each i , $A_i \subset S$. By the definition of union, $\cup A_i \subset S$. Hence, $\cup A_i \in \mathcal{B}$.
- Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be the two sigma algebras. (1) $\emptyset \in \mathcal{B}_1$ and $\emptyset \in \mathcal{B}_2$ since $\mathcal{B}_1$ and $\mathcal{B}_2$ are sigma algebras. Thus $\emptyset \in \mathcal{B}_1 \cap \mathcal{B}_2$. (2) If $A \in \mathcal{B}_1 \cap \mathcal{B}_2$, then $A \in \mathcal{B}_1$ and $A \in \mathcal{B}_2$. Since $\mathcal{B}_1$ and $\mathcal{B}_2$ are both sigma algebra $A^c \in \mathcal{B}_1$ and $A^c \in \mathcal{B}_2$. Therefore $A^c \in \mathcal{B}_1 \cap \mathcal{B}_2$. (3) If $A_1, A_2, \dots \in \mathcal{B}_1 \cap \mathcal{B}_2$, then $A_1, A_2, \dots \in \mathcal{B}_1$ and $A_1, A_2, \dots \in \mathcal{B}_2$. Therefore, since $\mathcal{B}_1$ and $\mathcal{B}_2$ are both sigma algebra, $\cup_{i=1}^{\infty} A_i \in \mathcal{B}_1$ and $\cup_{i=1}^{\infty} A_i \in \mathcal{B}_2$. Thus $\cup_{i=1}^{\infty} A_i \in \mathcal{B}_1 \cap \mathcal{B}_2$.

The union of two sigma algebras need not be a sigma algebra.

For a simple counter example consider

$$S = \{1, 2, 3\} \quad (\text{sample space})$$

$$A = \{\emptyset, \{1\}, \{2, 3\}, S\}$$

$$B = \{\emptyset, \{2\}, \{1, 3\}, S\}$$

It is not hard to show that A and B are both sigma algebras.

$$A \cup B = \{\emptyset, \{1\}, \{2\}, \{2, 3\}, \{1, 3\}, S\}$$

$$\{1\} \in A \cup B$$

$$\{2\} \in A \cup B$$

$$\{1\} \cup \{2\} = \{1, 2\} \notin A \cup B$$

So $A \cup B$ cannot be a sigma algebra.

1.4. a. "A or B or both" is $A \cup B$. From Theorem 1.2.9b we have $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

b. "A or B but not both" is $(A \cap B^c) \cup (B \cap A^c)$. Thus we have

$$\begin{aligned} P((A \cap B^c) \cup (B \cap A^c)) &= P(A \cap B^c) + P(B \cap A^c) && (\text{disjoint union}) \\ &= [P(A) - P(A \cap B)] + [P(B) - P(A \cap B)] && (\text{Theorem 1.2.9a}) \\ &= P(A) + P(B) - 2P(A \cap B). \end{aligned}$$

c. "At least one of A or B" is $A \cup B$. So we get the same answer as in a).

d. "At most one of A or B" is $(A \cap B)^c$, and $P((A \cap B)^c) = 1 - P(A \cap B)$.

Theorem 1.2.9 If P is a probability function and A and B are any sets in $\mathcal{B}$, then

- a. $P(B \cap A^c) = P(B) - P(A \cap B)$;
- b. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$;
- c. If $A \subset B$, then $P(A) \leq P(B)$.

Proof: To establish (a) note that for any sets A and B we have

$$B = \{B \cap A\} \cup \{B \cap A^c\},$$

and therefore

$$(1.2.6) \quad P(B) = P(\{B \cap A\} \cup \{B \cap A^c\}) = P(B \cap A) + P(B \cap A^c),$$

where the last equality in (1.2.6) follows from the fact that $B \cap A$ and $B \cap A^c$ are disjoint. Rearranging (1.2.6) gives (a).

To establish (b), we use the identity

$$(1.2.7) \quad A \cup B = A \cup \{B \cap A^c\}.$$

A Venn diagram will show why (1.2.7) holds, although a formal proof is not difficult (see Exercise 1.2). Using (1.2.7) and the fact that A and $B \cap A^c$ are disjoint (since A and A^c are), we have

$$(1.2.8) \quad P(A \cup B) = P(A) + P(B \cap A^c) = P(A) + P(B) - P(A \cap B)$$

from (a).

If $A \subset B$, then $A \cap B = A$. Therefore, using (a) we have

$$0 \leq P(B \cap A^c) = P(B) - P(A),$$

establishing (c). □

1.13 If A and B are disjoint, $P(A \cup B) = P(A) + P(B) = \frac{1}{3} + \frac{3}{4} = \frac{13}{12}$, which is impossible. More generally, if A and B are disjoint, then $A \subset B^c$ and $P(A) \leq P(B^c)$. But here $P(A) > P(B^c)$, so A and B cannot be disjoint.

1.7 a.

$$P(\text{scoring } i \text{ points}) = \begin{cases} 1 - \frac{\pi r^2}{A} & \text{if } i = 0 \\ \frac{\pi r^2}{A} \left[\frac{(6-i)^2 - (5-i)^2}{5^2} \right] & \text{if } i = 1, \dots, 5. \end{cases}$$

b.

$$\begin{aligned} P(\text{scoring } i \text{ points} | \text{board is hit}) &= \frac{P(\text{scoring } i \text{ points} \cap \text{board is hit})}{P(\text{board is hit})} \\ P(\text{board is hit}) &= \frac{\pi r^2}{A} \\ P(\text{scoring } i \text{ points} \cap \text{board is hit}) &= \frac{\pi r^2}{A} \left[\frac{(6-i)^2 - (5-i)^2}{5^2} \right] \quad i = 1, \dots, 5. \end{aligned}$$

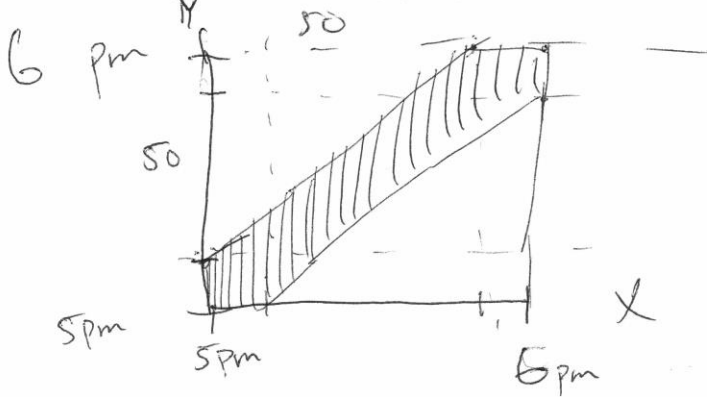
Therefore,

$$P(\text{scoring } i \text{ points} | \text{board is hit}) = \frac{(6-i)^2 - (5-i)^2}{5^2} \quad i = 1, \dots, 5$$

which is exactly the probability distribution of Example 1.2.7.

7. (5 Points) Two bankers each arrive at the station at some random time between 5PM and 6PM (arrival time for each of them is uniformly distributed). If they arrive before 5:50, they stay exactly 10 minutes and then leave. If a banker arrives after 5:50, they will leave at 6 PM. What is the probability that they will be at the station together on a given day?

Hint: Draw A Picture. The geometry of this problem is easier than the calculus.



$$\begin{aligned}
 P(\text{meet}) &= 1 - P(\text{don't meet}) \\
 &= 1 - \frac{\text{Area that White area}}{60^2} \\
 &= 1 - \frac{2 \cdot \frac{1}{2} \cdot \text{base} \cdot \text{height}}{60^2} \\
 &= 1 - \frac{2 \cdot \frac{1}{2} \cdot 50 \cdot 50}{60^2} = 1 - \left(\frac{50}{60} \right)^2 \\
 &= \frac{3600}{3600} - \frac{2500}{3600} \\
 &= \frac{1100}{3600} = \frac{11}{36}
 \end{aligned}$$

- 1.18 with n balls, n cells, and exactly one empty cell, we are looking for the event that there is 1 cell with zero balls, 1 cell with two balls, and $(n-2)$ cells with one ball each. Call this event A .

$$P(A) = \frac{\text{\# ways to get exactly 1 empty cell}}{\text{\# ways to ~~put~~ put } n \text{ balls into } n \text{ cells}}$$

Denominator: Since each of the n balls can be placed into any of the n cells, this can be done in n^n ways.

Numerator: There are 4 "tasks" to get exactly 1 empty cell

1. # ways to choose the empty cell: $= n$
2. # ways to choose the cell with two balls: $= n-1$
3. # ways to choose the two balls to go into the cell w/ 2 balls: $= \binom{n}{2}$
4. # ways to arrange the other $n-2$ balls into the last $n-2$ cells: $= (n-2)!$

$$P(A) = \frac{n(n-1)\binom{n}{2}(n-2)!}{n^n} = \boxed{\frac{\binom{n}{2}n!}{n^n}}$$

- 1.22 (a.) In order to have 180 days evenly distributed among the 12 months, there would be 15 days chosen in each month. The probability of the event of interest is:

$$\frac{\text{\# ways to choose 15 days in each month}}{\text{\# ways to choose 180 days out of 366}}$$

$$\frac{\begin{matrix} \text{Jan} & \text{Feb} & \text{Mar} & \text{Apr} & \text{May} & \text{Jun} & \text{Jul} & \text{Aug} & \text{Sep} & \text{Oct} & \text{Nov} & \text{Dec} \\ = \binom{31}{15} & \binom{29}{15} & \binom{31}{15} & \binom{30}{15} & \binom{31}{15} & \binom{30}{15} & \binom{31}{15} & \binom{31}{15} & \binom{30}{15} & \binom{31}{15} & \binom{30}{15} & \binom{31}{15} \end{matrix}}{\binom{366}{180}}$$

- (b.) The probability that the first day drawn is not from Sep is $\frac{336}{366}$ (since there are 30 days in Sep). After the first day is drawn, the probability the second day is not in Sept is $\frac{335}{365}$. Continuing in the same way we have:

$$\left(\frac{336}{366}\right) \cdot \left(\frac{335}{365}\right) \cdot \left(\frac{334}{364}\right) \cdot \dots \cdot \left(\frac{308}{338}\right) \left(\frac{307}{337}\right)$$

$$\frac{\left(\frac{336!}{306!}\right)}{\left(\frac{366!}{336!}\right)} = \frac{\left(\frac{336!}{30!306!}\right)}{\left(\frac{366!}{30!336!}\right)} = \frac{\binom{336}{30}}{\binom{366}{30}}$$

13.

ways to choose first rank $\downarrow$
 # ways to choose second rank $\downarrow$
 # way to choose 2 cards for pair $\downarrow$

$$\frac{(13)(12)(4)(2)(4)(3)}{\binom{52}{5}}$$
 # ways to choose 3 cards for triple $\downarrow$
 # ways to deal 5 cards $\uparrow$

1.34 a.

$$\begin{aligned}
 P(\text{Brown Hair}) &= P(\text{Brown Hair}|\text{Litter 1})P(\text{Litter 1}) + P(\text{Brown Hair}|\text{Litter 2})P(\text{Litter 2}) \\
 &= \left(\frac{2}{3}\right)\left(\frac{1}{2}\right) + \left(\frac{3}{5}\right)\left(\frac{1}{2}\right) = \frac{19}{30}.
 \end{aligned}$$

b. Use Bayes Theorem

$$P(\text{Litter 1}|\text{Brown Hair}) = \frac{P(\text{BH}|\text{L1})P(\text{L1})}{P(\text{BH}|\text{L1})P(\text{L1}) + P(\text{BH}|\text{L2})P(\text{L2})} = \frac{\left(\frac{2}{3}\right)\left(\frac{1}{2}\right)}{\frac{19}{30}} = \frac{10}{19}.$$

1.35 a. $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Since $P(\cdot)$ is a probability function, $P(A \cap B) \geq 0$ and $P(B) > 0$, so their ratio is ≥ 0 .

b. $P(\neg A|B) = \frac{P(\neg A \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$

c. If $A_1, A_2, \dots$ are disjoint

$$P\left(\bigcup_{i=1}^{\infty} A_i | B\right) = \frac{P\left(\bigcup_{i=1}^{\infty} A_i \cap B\right)}{P(B)} = \frac{P\left(\bigcup_{i=1}^{\infty} (A_i \cap B)\right)}{P(B)} = \sum_{i=1}^{\infty} \frac{P(A_i \cap B)}{P(B)} = \sum_{i=1}^{\infty} P(A_i|B)$$

1.36 (a.) the probability the target is hit at least twice is

$$\begin{aligned}
 1 - P(\text{target is hit 0 or 1 time}) &= 1 - P(\text{Hit 0 times}) - P(\text{Hit 1 time}) \\
 &= 1 - \left(\frac{4}{5}\right)^{10} - (10)\left(\frac{1}{5}\right)\left(\frac{4}{5}\right)^9 = \boxed{.624}
 \end{aligned}$$

(b.) Let 2 be the event the target is hit ^{at least} twice and let 1 be the event the target is hit at least once. We want to know $P(2|1)$. Note that since 2 is a subset of 1, $P(1 \cap 2) = P(2)$.

$$P(2|1) = \frac{P(2 \cap 1)}{P(1)} = \frac{P(2)}{P(1)} = \frac{1 - P(0 \text{ hits}) - P(1 \text{ hit})}{1 - P(0 \text{ hits})} = \frac{.624}{.893} = \boxed{0.699}$$

1.37

Three Prisoners

a.

Prisoner A's question: Which of B or C will be executed?

A = event that prisoner A is pardoned

B = event that prisoner B is pardoned

C = event that prisoner C is pardoned

W = event that Warden answers prisoner A's question with "Prisoner B."

X = place holder for any of the events A, B, or C in the following table.

If prisoner X is to be pardoned	then the Warden will answer A's question with either B or C	Use info from previous column to find $P(W X)$
A	$P(\text{says B}) = \gamma$ $P(\text{says C}) = 1 - \gamma$	$P(W A) = \gamma$
B	C	$P(W B) = 0$
C	B	$P(W C) = 1$

$$P(A|W) = \frac{P(A \cap W)}{P(W)} \rightarrow P(W|A)P(A) = \gamma \frac{1}{3}$$

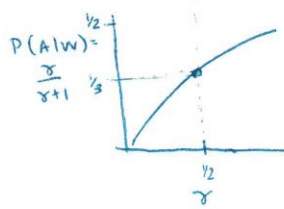
$$\begin{aligned} P(W) &\rightarrow P(W|A)P(A) + P(W|B)P(B) + P(W|C)P(C) \\ &= \gamma \left(\frac{1}{3}\right) + 0 \left(\frac{1}{3}\right) + (1) \left(\frac{1}{3}\right) \\ &= \frac{\gamma + 1}{3} \end{aligned}$$

$$\frac{\gamma \frac{1}{3}}{\frac{\gamma + 1}{3}} = \frac{\gamma}{\gamma + 1}$$

$$\text{if } \gamma = \frac{1}{2} \text{ then } P(A|W) = \frac{1}{3}$$

$$\text{if } \gamma < \frac{1}{2} \text{ then } P(A|W) < \frac{1}{3}$$

$$\text{if } \gamma > \frac{1}{2} \text{ then } P(A|W) > \frac{1}{3}$$



b. we can find $P(C|W) = \frac{(\frac{1}{3})}{\frac{1}{2}} = \frac{2}{3}$ So if A switches fates with C, P(A is pardoned) does increase to $\frac{2}{3}$